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Inference

1. Whatever specific percentage of the survey participants said they preferred Product X, the percentage of individuals with that preference in a much larger population might be exactly the same or much different. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined from (1) alone; NOT sufficient.
2. The statement that a small sample may not be representative of a population's preferences suggests that the sample also may be representative of that population's preferences. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined from (2) alone; NOT sufficient.

If (1) and (2) are both true, it is possible that 70% of the broader population prefers Product X, just as 70% of the survey respondents claimed to prefer it, but it's also possible that some much different percentage of the broader population prefers Product X. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined even from both statements together.

The correct answer is E; both statements together are still not sufficient.

•

What is a limiting factor when it comes to us using the most effective solution?

1. We have identified several potential solutions, but a lack of funding hinders their implementation.
2. As the most effective solution requires extensive resources, we must opt for a more affordable option given our budget.

Inference

1. Even if a lack of funding hinders the implementation of several potential solutions we have identified, maybe none of those potential solutions is the most effective solution. Perhaps the most effective solution is well within our budget, but we haven't identified it. Thus, an example of a limiting factor when it comes to us using the most effective solution cannot be determined from (1) alone; NOT sufficient.
2. If the extensive resources required by the most effective solution require us to opt for a more affordable option given our budget, then cost must be one limiting factor when it comes to us using the most effective solution. Thus, (2) alone provides an example of a limiting factor when it comes to us using the most effective solution; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

What are all the important considerations for making accurate projections?

1. Based on the available data, we believe that the market demand for our product will increase by 20% in the next quarter.
2. Without comprehensive market research, our projection may not account for external factors that could influence fluctuations in demand.

Inference

1. An observation about how "we" believe market demand for a product will change in the next quarter provides no information about important considerations for making accurate projections. Thus, what considerations are important for making accurate projections cannot be determined from (1) alone; NOT sufficient.
2. (2)'s claim that our projection may not account for external factors without comprehensive market research doesn't tell us what considerations are generally important for making accurate projections. (2) leaves open the possibility that our projection may account for external factors without comprehensive market research. It also leaves open the possibility that the projection may be accurate even if it doesn't account for external factors, or that it may be inaccurate even if it does account for such factors. Furthermore, even if comprehensive market research were one important consideration for making our projection, that wouldn't tell us what the other important considerations for making the projection are. And finally, even if we knew all the important considerations for making this particular projection, that wouldn't tell us what considerations are important for making accurate projections in general. Thus, what considerations are important for making accurate projections cannot be determined from (2) alone; NOT sufficient.

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cannot be determined from (2) alone; NOT sufficient.

Since (1) is completely irrelevant to determining the considerations important for making accurate projections, while (2) is inadequate for determining them, even both statements together can't tell us what considerations are important for making accurate projections.

The correct answer is E; both statements together are still not sufficient.

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Does Company X strive to maintain a low defect rate in order to ensure product quality?

1. Company X's manufacturing process has a defect rate of 2% according to quality control data.
2. Monitoring the defect rate in a manufacturing process is essential to ensure product quality, and a low percentage indicates a well-maintained production process.

Inference

1. Knowing that quality control data says the defect rate for Company's X's manufacturing process is 2% tells us nothing about whether Company X strives to maintain a low defect rate. First, the quality control data may be inaccurate or fraudulent. Second, without further context, we can't tell whether a 2% defect rate is low or high. And finally, even if a 2% defect rate is low, we don't know whether Company X is striving to keep it low to ensure product quality. Thus, whether Company X strives to maintain a low defect rate in order to ensure product quality cannot be determined from (1) alone; NOT sufficient.
2. Even if monitoring the defect rate in a manufacturing process is essential to ensure product quality, Company X may not monitor it. And even if a low defect rate indicates a well-maintained production process, Company X's defect rate may be high. Thus, whether Company X strives to maintain a low defect rate in order to ensure product quality cannot be determined from (2) alone; NOT sufficient.

Even if (1) is true, the real defect rate for Company X's manufacturing process might be higher than the quality control data suggests. And even if the defect rate is really 2%, that might be quite a high rate for Company X's product and manufacturing process. If the defect rate is high, then (2) does not imply that Company X is striving to maintain a low defect rate. Thus, (1) and (2) together don't tell us whether Company X strives to maintain a low defect rate.

The correct answer is E; both statements together are still not sufficient.

•

How might career advancement be described as potentially linked to both growth and decline?

1. Working long hours and overtime may lead to higher productivity and career advancement but is likely to cause a decline in physical and mental well-being.
2. Maintaining a work-life balance is important to reduce the negative effect that working too much has on a person's quality of life.

Inference

1. (1) describes how working long hours and overtime is potentially linked to a type of growth (in productivity) and also to a type of decline (in well-being) as well as to career advancement. That is, (1) directly describes a way that career advancement is potentially linked to both growth and decline, via working long hours and overtime. Since (1) itself is an answer to the question, it is sufficient to answer it; SUFFICIENT.
2. (2) says nothing about growth or career advancement. Thus, (2) alone doesn't tell us how career advancement might be described as potentially linked to both growth and decline. NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

What is an example of how investment in healthcare resources might sometimes create a challenge?

1. Investment in both advanced medical technology and specialized treatments can improve patient outcomes for certain conditions. Nevertheless, it may put a strain on healthcare budgets.
2. When it comes to allocating resources in the healthcare sector, directing money toward the purchase of advanced medical equipment could potentially limit access to other essential services and treatments.

Inference

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1. (1) gives an example of how investment in certain healthcare resources (advanced medical technology and specialized treatments) might create a challenge (by straining healthcare budgets). Thus, (1) alone directly answers the question; SUFFICIENT.
2. (2) gives an example of how investment in certain healthcare resources (advanced medical equipment) might create a challenge (by limiting access to other essential services and treatments). Thus, (2) alone directly answers the question; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

A person bought x number of apples, y number of oranges, and z number of mangoes from a shop. What is the total price of this transaction if each apple costs $2, each orange costs $2.50, and each mango costs $3?

1. The number of apples bought is 12 more than the number of oranges bought. The number of mangoes bought is half the number of apples bought.
2. The average number of fruits purchased from the three varieties is greater than 29 and less than 33.

Applied Problems

1. (1) tells us that $x = y + 12 = 2z$. But this doesn't tell us exactly how much fruit was purchased. For example, it could be that x is 14, y is 2, and z is 7. Or it could be that x is 16, y is 4, and z is 8. The total price would be greater in the latter case, because more fruit of each type would have been purchased. Thus, the total price of the transaction cannot be determined from (1) alone; NOT sufficient.
2. (2) tells us only a range of values for the average number of fruits purchased of the three varieties. Even if we knew more precisely that the average was exactly 30, we still couldn't tell the total price of the transaction, because the different types of fruit differ in price. For example, this average would be 30 if the transaction included 10 apples, 30 oranges, and 50 mangoes. But the average would also be 30 if the transaction included exactly 30 apples, 30 oranges, and 30 mangoes. The total price would be higher in the former case than in the latter, because the mangoes cost more apiece than the apples. Thus, the total price of the transaction cannot be determined from (2) alone; NOT sufficient.

If (1) and (2) are both true, then we know from (1) that $x = y + 12 = 2z$. And we know from (2) that the average number of fruits of the three varieties is greater than 29 and less than 33; that is, $3(29) = 87 < x + y + z < 3(33) = 99$. But this still doesn't tell us exactly how much fruit was purchased. Notice that the average need not be a whole number. It could be a fraction. For example, it's possible that x is 40, y is 28, and z is 20. That gives us $x + y + z = 88$, yielding an average of 88/3 = 29 1/3—which is greater than 29 and less than 33. Or it could be that x is 42, y is 30, and z is 21, which gives us $x + y + z = 93$, yielding an average of 93/3 = 31—which is also greater than 29 and less than 33. The total price would be greater in the latter case, because more fruit of each type would have been purchased. Thus, the total price of the transaction cannot be determined even from the two statements together.

The correct answer is E; both statements together are still not sufficient.

•

Adam received a certain monetary amount in bills having only three denominations: $1, $5, and $10. What is the total monetary amount in bills that he received?

1. The numbers of $1, $5, and $10 bills Adam received, respectively, are in the ratio 3:5:8.
2. The ratio of the monetary amount Adam received in $1 bills to the monetary amount he received in $10 bills is 3:80.

Applied Problems

1. More than one total monetary amount in bills is possible that satisfies the given information and (1). For example, the numbers of $1, $5, and $10 bills, respectively, could be 3, 5, and 8 for a total monetary amount of $108. Or the numbers of $1, $5, and $10 bills, respectively, could be 6, 10, and 16 for a total monetary amount of $216; NOT sufficient.
2. Since no information is given about the number of $5 bills, the total monetary amount in bills cannot be determined; NOT sufficient.

Letting k be the proportionality constant in (1), it follows that the numbers of $1, $5, and $10 bills, respectively, are

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Letting k be the proportionality constant in (1), it follows that the numbers of \$1, \$5, and \$10 bills, respectively, are $3k$, $5k$, and $8k$. Hence, the total monetary amount Adam received in \$1 bills is $3k(\$1) = \$3k$, and the total monetary amount he received in \$10 bills is $8k(\$10) = \$80k$. Thus, (1) implies that the ratio of the monetary amount Adam received in \$1 bills to the monetary amount he received in \$10 bills is $3k$:$80k$, or 3:80. Since this result is identical to the information provided by (2), it follows that taking (1) and (2) together provides no more information than (1) alone, which is not sufficient. Therefore, (1) and (2) together are not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

Exactly 3 investors—A, B, and C—invested in a certain business, each for a certain number of months. For these investments, what is the ratio of the profit earned by A to the profit earned by B to the profit earned by C? (Note: Profit earned is proportional to the product of investment amount and investment duration.)

1. A invested twice as much as B, and C invested half as much as A and B together.
2. All 3 investments were for the same number of months.

Applied Problems

Let x, y, and z be the numbers of months, respectively, that A, B, and C invested. Let $\$a$, $\$b$, and $\$c$ be the amounts, respectively, that A, B, and C invested. Then, if k is the constant of proportionality associated with these investments, the profits earned by A, B, and C, respectively, are $\$kax$, $\$kby$, and $\$kcz$. We are to determine the ratio kax:kby:kcz. Equivalently, we are to determine the ratio ax:by:cz.

1. We are given that $a = 2b$ and $c = \frac{1}{2}(a+b)$. Since we are given no information about x, y, or z, it is not possible to determine the ratio ax:by:cz; NOT sufficient.
2. We are given that $x = y = z$. Since we are given no information about a, b, or c, it is not possible to determine the ratio ax:by:cz; NOT sufficient.

From (2) we have ax:by:$cz = ax$:bx:$cx = a$:b:c. From (1) we have $a = 2b$ and $c = \frac{1}{2}a + \frac{1}{2}b$, and hence $c = \frac{1}{2}(2b) + \frac{1}{2}b, = b$. Therefore, a:b:$c = 2b$:b:$\frac{1}{2}b = 2$:1:$\frac{1}{2}$, or 4:2:3.

The correct answer is C; both statements together are sufficient.

•

The ratio of A's current age to B's current age is 5:8. What will be the sum of their ages 10 years from now? (Note: Assume that neither A nor B dies in the next 10 years.)

1. If A had been born 2 years later and B had been born 2 years earlier, then the ratio of A's current age to B's current age would be 14:25.
2. A's age 5 years ago was half the age that B will be 2 years from now.

Arithmetic Applied Problems

Let k be the constant of proportionality for the 5:8 ratio of A's current age to B's current age. Then A's current age is $5k$ and B's current age is $8k$. We are to determine the value of $(5k+10)+(8k+10) = 13k+20$, or equivalently, we are to determine the value of k.

1. If A had been born 2 years later, then A's current age would be $5k-2$, and if B had been born 2 years earlier, then B's current age would be $8k+2$. We are told that the ratio $(5k-2)$:$(8k+2)$ is equal to 14:25. Thus, $\frac{(5k-2)}{(8k+2)} = \frac{14}{25}$. Cross multiplying gives a first-degree equation that can be solved for a unique value of k. Although it is not necessary to find the value of k, solving this equation gives $k = 6$; SUFFICIENT.
2. We are given that $5k - 5 = \frac{1}{2}(8k+2)$, which can be solved for a unique value of k. Although it is not necessary to find the value of k, solving this equation gives $k = 6$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

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•

A's father is 6 years older than her mother. The ratio of A's age to her mother's age is 1:4. What is the age of A's father?

1. A's father is 30 years older than A.
2. The ratio of A's age to her father's age is 4:19.

Arithmetic Applied Problems

Let a, f, and m be, respectively, the ages of A, A's father, and A's mother. We are given that $f = m + 6$ and $\frac{a}{m} = \frac{1}{4}$. Thus, $m = f - 6$, and hence we get $\frac{a}{(f-6)} = \frac{1}{4}$, or $4a = f - 6$. We are to determine the value of $f = 4a + 6$.

1. We are given that $f = 30 + a$. Since we also know that $f = 4a + 6$, it follows that $30 + a = 4a + 6$. This is a first-degree equation that can be solved for a unique value of a, which can then be used to determine a unique value of $f = 4a + 6$. Although it is not necessary to find these values, solving gives $a = 8$ and $f = 38$; SUFFICIENT.
2. We are given that $\frac{a}{f} = \frac{4}{19}$. Since we also know that $f = 4a + 6$, it follows that $\frac{a}{(4a+6)} = \frac{4}{19}$. Cross multiplying gives a first-degree equation that can be solved for a unique value of a, which can then be used to determine a unique value of f. Although it is not necessary to find these values, solving gives $a = 8$ and $f = 38$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

The salary of A is what percentage of the combined salaries of B and C?

1. Twice the salary of A is equal to three times the salary of B.
2. 150% of B's salary is \$160 more than 120% of C's salary.

Algebra Applied Problems

Let $\$a$, $\$b$, and $\$c$ be the salaries, respectively, of A, B, and C. We are to determine a as a percent of $b + c$, or equivalently, we are to determine the value of $\frac{a}{b+c}$.

1. We are given that $2a = 3b$, or $a = \frac{3}{2}b$. Since we are given no information about the value of the positive number c, it follows that the value of $\frac{a}{b+c}$ cannot be determined; NOT sufficient.
2. We are given that $1.5b = 160 + 1.2c$. Since we are given no information about the value of the positive number a, it follows that the value of $\frac{a}{b+c}$ cannot be determined; NOT sufficient.

From (1) we have $\frac{a}{b+c} = \frac{\frac{3}{2}b}{b+c}$. From (2) we have $15b = 1{,}600 + 12c$, or $12c = 15b - 1{,}600$. To avoid additional complications with fractions when incorporating $12c = 15b - 1{,}600$ into $\frac{\frac{3}{2}b}{b+c}$, multiply the numerator and denominator of this last fraction by 12 to get $\frac{18b}{12b+12c}$. Now replace $12c$ in this fraction with $15b - 1{,}600$ to get $\frac{18b}{12b+15b-1{,}600} = \frac{18b}{27b-1{,}600}$. This last fraction is equal to the value of $\frac{a}{b+c}$, and it is clear that the value of this last fraction can vary. For example, if b is very large, then $\frac{18b}{27b-1{,}600} \approx \frac{18b}{27b} = \frac{2}{3}$; and if $27b$ is

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value of this last fraction can vary. For example, if b is very large, then $\frac{18b}{27b-1{,}600} \sim \frac{18b}{27b} = \frac{2}{3}$, and if $27b$ is only a little larger than 1,600, then $\frac{18b}{27b-1{,}600} = \frac{18b}{\text{small positive number}}$ will be much larger than $\frac{2}{3}$ (e.g., if $b = 60$, then the value of this fraction is 54). Therefore, (1) and (2) together are NOT sufficient.

The correct answer is E; both statements together are still not sufficient.

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